PRACTICAL- 14

Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

AIM / Problem Definition: Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

Tools / Technology Used: PHP 8.x, MySQL, Apache / XAMPP, Modern CSS Grid & Flexbox, SVG / Canvas Analytics Charts, VS Code

Project Implementation & Architecture

Project Overview (StudentHub — Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log)

Practical 14 develops the high-level System Administrator Operations Hub for the StudentHub Portal (admin.php). It establishes strict Role-Based Access Control (RBAC), aggregates live institutional metrics directly from MySQL, features an interactive User Account Status Toggle, provides an SVG monthly registrations analytics chart, and maintains a comprehensive system audit trail.

Technical Architecture & Workflow

Strict RBAC Route Protection: Guarded by requireRole("admin"). Any unauthenticated visitor is redirected to login, while authenticated students receive an HTTP 403 Forbidden access-denied screen.

Dynamic Role-Based Menu System: Navigation menus render dynamically based on user privilege tiers (administrators access management and moderation tools; students access personal dashboards and event registrations).

Live Relational Metric Aggregation: Executes optimized SQL COUNT(*) aggregation queries calculating total enrolled students, active campus events, registered user accounts, and confirmed event registrations.

Intermediate Extension: User Account Status Toggle: Allows administrators to instantly activate or deactivate user accounts with automated audit logging in audit_logs and self-deactivation prevention.

Advanced Extension: Monthly Event Registrations Analytics Chart: Queries registrations using SQL DATE_FORMAT(registration_date, "%b %Y") and GROUP BY, rendering an interactive bar and trend chart with tooltips, monthly values, and peak participation statistics.

Real-Time Administrative Audit Trail: Maintains a live tabular audit log stream capturing administrative events, timestamps, executing usernames, action types, details, and client IP addresses.

Output Verification & Functional Screens

Output 1: Administrator Operations Hub Dashboard

Comprehensive control center rendering dynamic metric cards (Total Students, Active Events, Registered Users, Event Participations) and quick-action module shortcuts.

Output 2: Monthly Event Registrations Analytics Chart (Advanced Extension)

Interactive SVG bar chart displaying monthly student event participation trends across calendar months with proportional heights and data tooltips.

Output 3: User Account Management & Status Toggle Interface (Intermediate Extension)

Table listing user accounts with roles (Admin/Student), status badges (Active/Inactive), and interactive Deactivate/Activate buttons with confirmation.

Output 4: User Deactivation Success & Audit Logging

UI feedback alert confirming account status toggle alongside verified insertion of USER_STATUS_TOGGLED entry in audit_logs table.

Output 5: Latest Event Registrations Table

Live table rendering recent student event signups with student names, roll numbers, event titles, scheduled dates, and attendance status.

Output 6: Real-Time Administrative Audit Trail Stream

Audit logs table detailing chronological administrative activity with timestamps, usernames, action codes, and IP addresses.

Key Questions & Technical Analysis

Q1. Are dashboard values loaded dynamically from the database?

Yes. All statistical indicators (student count, event count, user accounts, registrations) are calculated in real time using SQL COUNT(*) queries against live database tables, ensuring zero hardcoded figures.

Q2. Is access restricted to admin users?

Yes. The dashboard strictly enforces requireRole("admin"). If a session lacks admin privileges, execution halts immediately with HTTP 403 Forbidden and an access-denied error template.

Q3. Are role-based menus working?

Yes. The navigation bar inspects session role credentials and renders tailored links: administrators receive administrative and moderation modules, whereas students receive personal academic links.

Q4. Are important admin actions recorded in audit_logs?

Yes. Critical administrative actions—including event creation, student updates, user deactivations, and deletions—automatically insert records into the audit_logs table with user_id, action name, descriptive details, and client IP address.

GitHub Repository Link

GitHub Repository URL: https://github.com/smit-2406/studenthub-website

Conclusion & Learning Outcomes

Practical 14 delivered an enterprise-grade administrative governance dashboard featuring real-time SQL aggregation, dynamic role-based navigation, interactive user account authorization toggles, visual event analytics, and complete audit trail traceability. This fulfills Course Outcomes CO1, CO4, and CO5.